Siva G Nair

AI Engineer | Generative AI • Computer Vision • NLP
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Professional Summary

AI Engineer specializing in Generative AI, Computer Vision, and NLP, with hands-on experience building and deploying industrial and enterprise AI systems. Engineered RAG and LLM-based document automation workflows, PyTorch/OpenCV visual inspection systems, and AWS data pipelines, delivering 95% defect detection accuracy, 90% document classification accuracy, 40% improved information retrieval, and processing 1M+ records. Proficient in Python, SQL, PyTorch, TensorFlow/Keras, OpenCV, RAG, LLMs, embeddings, vector databases, and cloud-based AI deployment.

Professional Experience

AI Engineer | **Ombrulla** | Kochi, Kerala | Dec 2024 – May 2025

- Designed and deployed industrial AI proof-of-concepts for computer vision and NLP, incorporating LLM-based document processing and achieving 90% automated document classification accuracy, reducing client feedback cycles by 30%
- Engineered a RAG pipeline for enterprise document question answering, tabular-data interaction, and automated summarization, improving information retrieval efficiency by 40%
- Integrated camera systems with PyTorch CNNs and OpenCV for visual inspection, enabling local model training and deployment with 95% defect detection accuracy
- Streamlined Python, Pandas, and NumPy data pipelines on AWS for real-time processing of 1M+ records, supporting model training and inference

Data Science Intern | **Luminar Technolab** | Kochi, Kerala | May 2024 – Nov 2024

- Developed and optimized predictive analytics models with Scikit-learn, TensorFlow, and Keras, improving time-series forecasting accuracy by 25%
- Preprocessed, cleaned, and feature-engineered datasets exceeding 500K records using Pandas and NumPy to improve model-training inputs
- Created stakeholder-facing dashboards and visualizations with Power BI, Matplotlib, and Seaborn, reducing decision-making time by 20%

Projects

RAG-Based Email Automation Agent

GenAI • OCR • Vector Search • Ollama • Milvus

- Engineered a multimodal RAG workflow integrating EasyOCR, PyTorch, Ollama, and Milvus, reducing manual processing time by 40%
- Built a pipeline that fetches emails, preprocesses content and attachments (OCR for images), and generates context-aware responses; added event detection, notification triggers, and a local Ollama LLM for private analysis

RAG-Based Text-to-SQL Assistant

RAG • LLM • PostgreSQL

- Engineered a RAG framework that gives an LLM database schema and context to translate plain-English questions into PostgreSQL queries and retrieve relevant records

Botanic Vision – Rice Plant Disease Detection App

Flask • TensorFlow • Computer Vision

- Developed a Flask/TensorFlow web app identifying five major rice plant diseases with 93% accuracy

Text Emotion Detection

NLP • Keras • Streamlit

- Built a Streamlit NLP application for five-class emotion classification with 85% accuracy using Python, NLTK, and Keras

Personalised Nutrition System Using LLMs

LLM • Personalised Nutrition • Recipe Recommendation

- Recommends recipes from ingredients users already have to reduce food waste, ranking options by dietary goals, health preferences, and declared allergies, and estimating calories for each suggestion

GymFit – AI Gym and Calorie Tracker
LLM • Mobile App • Calorie Tracking

- Built a fitness app for gym training, daily calorie intake, and three goals: fat reduction, maintenance, and muscle gain
- Calculates BMI, maintenance calories, and a personalised daily calorie target from age, height, weight, and activity level
- Uses an LLM to generate meal schedules with calories and protein, give training tips, and estimate calories from plain-text food descriptions; stores daily weight entries in the cloud with a progress graph

Power BI Dashboards – Twitch Analytics and HR Attrition
Power BI • Power Query • DAX

- Built a real-time Twitch viewership dashboard and an HR attrition dashboard using Power Query and DAX measures, analyzing turnover by department, age, salary range, and job satisfaction

Technical Skills

Languages: Python, SQL, JavaScript, Java
Machine Learning: PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, Hugging Face
Generative AI: RAG, LLMs, Prompt Engineering, Embeddings, AI Agents, MCP
LLM Frameworks: LangChain, LlamaIndex, Ollama, EasyOCR
Databases: PostgreSQL, MongoDB, Milvus, Vector Databases
Data and Analytics: Pandas, NumPy, Power BI, Power Query, DAX, Matplotlib, Seaborn
Cloud and Deployment: AWS, Docker, Flask, Streamlit, Heroku, Cloudflare
Tools: Git, GitHub

Education

Bachelor of Technology in Computer Science | Sree Buddha College of Engineering | 2020 – 2024
Specialization: Artificial Intelligence and Machine Learning | APJ Abdul Kalam Technological University